The Frohmanian Tethered Metriplectic World Model in Lean 4: Canonical Bracket Splitting, Encoding Obstructions, and the Newton Potential

Companion paper to `BenFrohman/metriplectic-world-model`

Benjamin Stanley Frohman
ORCID 0009-0006-7068-3718
`frohmanbenjamin@gmail.com`

Draft, 31 October 2026 (repository HEAD `5d208d9`)

Draft — tracks build in progress. This repository formalizes a *model*: the tethered metriplectic system, which adds a symmetric tether bracket and cubic damping of peak vorticity to 3D vorticity dynamics. It proves nothing about unmodified Navier–Stokes ($\kappa = 0$) and does not resolve the Clay Millennium problem.

Abstract

We report a Lean 4/Mathlib formalization (0 `sorry`; axioms only `propext`, `Classical.choice`, `Quot.sound`) of the author's tethered metriplectic model. Proved unconditionally: the unique antisymmetric/symmetric splitting of a bilinear bracket (`split_exists_unique`); a non-circular canonicity theorem forcing the metric bracket to the form $c\langle \Pi_u x, \Pi_u y\rangle$ (`canonical_metric_bracket`); a uniform vorticity ceiling for the comparison inequality $M' \le CM^2 - \kappa M^3$; a log-loss replacement for the stretching hypothesis giving a ceiling when $\kappa > 4C_{\mathrm{abs}}$; several encoding counterexamples; and the first parts of the Newton-potential identity on $\mathbb{R}^3$. Conditional and open items are listed by their exact Prop names.

1 Naming

`Pi_u` is the author's own name (manuscript of 31 May 2026): the projection $\Pi_u x = x - (\langle u, x\rangle / \|u\|^2)u$ off $\mathrm{span}\{u\}$. All other identifiers defined in this repository are *coined in this build* (marked [coined]), including `canonical_metric_bracket`, kept by the author's choice and flagged as coined. Modules under `FrohmanianSymplecticTether/` and `TetherMayForm/` are ported verbatim from earlier repositories.

2 Manuscript framing

The May/June 2026 manuscript introduces a tethered Lyapunov functional S_ε and reduces regularity to the comparison ODE $y' = Cy^2 - \kappa'' y^3$ for the peak vorticity, through transport cancellation, viscous dissipation at the maximum, a Hölder–Sobolev gate, and Young absorption at $\varepsilon = \kappa/4$ (Proposition 2). Within the model, the cubic term comes from the tether. The manuscript's hypothesis (H5), a non-circular majorant for the vorticity, is the genuine gap; for $\kappa = 0$ it is the Clay core. Here κ has units of time and breaks Navier–Stokes scaling, which is why the model is distinct from NS.

3 Brackets and canonicity (`Bracket.lean`, `Canonical.lean`)

Theorem 1. *`split_exists_unique`[coined]: every bilinear bracket is uniquely an antisymmetric part plus a symmetric part.*

Theorem 2. *`canonical_metric_bracket`[coined]: on a real inner product space, a bilinear form B that is symmetric (C1), degenerate along u (C2: $B(u,\cdot)=0$) and invariant under every linear isometry fixing u equals $c\,\langle \Pi_u x, \Pi_u y\rangle$. The hypotheses do not mention Π_u. `canonical_metric_bracket_sign`[coined]: if $B(x,x)\le 0$ then $c\le 0$. `Pi_u_form_symm_degen`[coined]: the forced form satisfies C1, C2.*

Scope: pointwise at one state; does not force the $|\omega|^2$ weight, the value of κ, or C3. Also proved: `energy_conserved`, `energy_const_along`, `entropy_production`, `gen_entropy_eq`, `entropy_monotone_along`[coi

4 Ceiling and stretching (`Ceiling.lean`, `AuditFixes.lean`)

Theorem 3. *`uniform_ceiling`[coined]: if $M' \le CM^2 - \kappa M^3$, $\kappa > 0$, then $M(t) \le \max(M(0), \max(C/\kappa, 0))$ for $t \ge 0$. Also `absorbing`, `ceiling_antitone_kappa`, `kappa_zero_no_ceiling`[coined] (at $\kappa = 0$, $1/(1-t)$ satisfies $M' \le M^2$ and blows up).*

Correction. The hypothesis `StretchingCZ` (a pure y^3 bound on stretching at the maximum) is *false as a sup-norm bound* for Navier–Stokes: the known bound carries a logarithmic loss. Results using it are relabeled as model-only. Its honest replacement is `StretchingLogLoss` $d\,C_1\,C_2$[coined]: $St \le C_1 y^3 + C_2 y^3 \log(1+y)$.

Theorem 4. *`maxpoint_rate_logLoss`[coined]: under `EnergyIdentity`, `StretchingLogLoss`, `ViscousAtMax`, `TetherFeedback`, `CrossControlled`,*

$$y' \le \left(C_1 + \tfrac{\kappa}{4} + \tfrac{4C_2^2}{\kappa}\right) y^2 - \left(\tfrac{\kappa}{4} - C_{\text{abs}}\right) y^3,$$

so with `uniform_ceiling` a uniform ceiling follows whenever $\kappa > 4C_{\text{abs}}$ (Young absorption at $\varepsilon = \kappa/4$).

5 Encoding obstructions (`ChoiceFree`, `HelmholtzObstruction`, `TestClass`)

- `not_streamFunctionRepModGrad`[coined]: `StreamFunctionRepModGrad` fails for every ω — representatives are determined only a.e.; uniqueness must be stated almost everywhere (open).
- `arnoldBracketOf_antisymm`, `entropyCasimir_arnold`[coined]: a choice-free Arnold bracket and a Casimir criterion.
- `not_curl_biotSavart_eq`[coined]: curl(BiotSavart ω) $= \omega$ fails for the point spike in the current encoding; `helicity_casimir_of_curlBiotSavart`[coined] is conditional on `CurlBiotSavartOn`.
- `IsTestOrbit`[coined] (smooth, compactly supported, div-free); `not_isTestOrbit_pointSpike`, `helicity_casimir_test`[coined] (conditional on `CurlBiotSavartTest`); `integrableOn_ball_norm_rpow_neg`

 © 2026 Benjamin Stanley Frohman. Apache-2.0.

6 Newton potential on $\mathbb{R}^3$ (`NewtonKernel`, `RegularizedKernel`)

Proved: `scalarLaplacian_newtonRadial` ($\Delta(1/|y|) = 0$, $y \neq 0$), `contDiff_newtonPotential`, `hasFDerivAt_newtonPotential` (differentiation under the integral); for $N_\varepsilon = (|y|^2 + \varepsilon^2)^{-1/2}$: `scalarLaplacian_regNewton` ($\Delta N_\varepsilon = -3\varepsilon^2(|y|^2 + \varepsilon^2)^{-5/2}$), `regNewton_le`, `tendsto_regNewton`, `integral_neg_scalarLaplacian_regNewton` ($\int -\Delta N_\varepsilon = 4\pi$), `tendsto_regPeak_approxIdentity` (all [coined]). The regularized route avoids spherical Green identities, which Mathlib lacks. *Open:* $N_\varepsilon \star \Delta g = (\Delta N_\varepsilon) \star g$; $\varepsilon \to 0$ limit; $\Delta(N \star g) = N \star \Delta g$; hence $\Delta(N \star g) = -4\pi g$; then $\operatorname{curl}\operatorname{curl} = \nabla \operatorname{div} - \Delta$ and identification with `BiotSavart`.

7 Status

- **Proved unconditionally:** `split_exists_unique`, `canonical_metric_bracket` (+ sign, symm/degen), energy/entropy structure, `uniform_ceiling`, `maxpoint_rate_logLoss` (as an implication from its listed Props), the obstructions above, Newton steps 1–2 and partial step 3, the scalar instance `instance_global_regularity`.
- **Conditional on named Props:** `model_uniform_ceiling` (`TetheredMaxRate`); `model_global_regularity` (`TetheredMaxRate`, `ModelContinuation`); `model_global_regularity_of_realization` (`MaxPointRealiza` whose fields include `StretchingCZ` — false as an L^∞ NS bound); velocity-tether energy/entropy (`KineticEnergyCurlIdentity`, `EntropyCasimir`); helicity Casimir (`CurlBiotSavartTest`).
- **Open:** `CurlBiotSavartTest`; a.e. uniqueness modulo gradients (`StreamFunctionRepModGradOn`); `EntropyCasimir` for the concrete bracket; deriving the $|\omega|^2$ weight, κ and C3 from C1–C3; the link of maximizer data to an actual vorticity field; and, for unmodified NS, the (H5) majorant, i.e. the Clay core. Neither the Clay problem nor NS regularity is claimed.
- The type named `T3` is $\mathbb{R}^3$, not the torus.

License

Copyright 2026 Benjamin Stanley Frohman. Licensed under the Apache License, Version 2.0.

 © 2026 Benjamin Stanley Frohman. Apache-2.0.